Francesco Calice

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Education

MSc Quantum Fields and Fundamental Forces — Imperial College London, UK
Theoretical Physics — *Sep. 2026 – Aug. 2027*

- Elective Modules: Differential Geometry, Supersymmetry, Beyond the Standard Model, String Theory

BSc Physics with Theoretical Physics | 1st Class Honours — Imperial College London, UK
Sep. 2023 – Jun. 2026

- **84.98%** final year average - **Dean's List Prize** for top 10% performance.
- Ranked **2/235** in Physics Lab and Analysis Module.
- Relevant Modules: Group Theory (92%), Advanced Classical Physics (92%), Quantum Foundations (90%).

Research Experience

CERN – Neutrino Group — Jun. 2026 – Aug. 2026
Summer Student | EP-Nu | Supervisor: Dr Animesh Chatterjee — *Geneva, Switzerland*

- Selected for the CERN Summer Student programme from $\sim$ **20,000 applicants**; $\sim$ 20 UK students admitted.
- Conducted a BSM phenomenology analysis of heavy neutral leptons at ProtoDUNE-SP, developing a new neutrino background-rejection algorithm and deriving new **background-inclusive sensitivity projections** for active-sterile mixing U^2_{e4} and $U^2_{\mu 4}$.
- Presented findings to CERN's EP-Nu research group and authored a publicly available CERN research report documenting the analysis, validation and projected constraints.

Imperial College London – Dark Matter Direct Detection — Oct. 2025 – Jan. 2026
Bachelor's Research Project | Supervisor: Prof Stefan Söldner-Rembold — *London, UK*

- Modelled spin-independent WIMP–nucleon interactions, deriving analytic cross-sections from first principles.
- Developed a toy model to estimate detector sensitivity limits and investigate the impact of detector parameters.
- Presented and defended the theoretical derivations and computational methodology in an internal viva; **awarded highest mark in the year (86%).**

Imperial College London – Type Ia Supernovae — Jan. 2025 – Mar. 2025
Undergraduate Researcher | Supervisor: Dr Eleni Tsaprazi — *London, UK*

- Built an SN Ia rate-modelling pipeline in Python, analysing $\sim$ **150M simulated galaxies** from SQL catalogues.
- Implemented Star-Formation-History models, using Poisson sampling to analyse rate–morphology relationships.
- Presented findings at an international cosmology teleconference; incorporated into a subsequent grant proposal.

Independent Projects

Symmetry and Interpretability in NPEs for Neutrino Physics — Aug. 2026 – Present
Independent Project

- Investigating how neural networks learn and represent **approximate physical symmetries**, using degeneracies in neutrino oscillation parameters as a controlled test case.
- Developing symmetry-aware neural posterior estimators with normalising flows to study multimodal inference and symmetry breaking.
- Developing framework to probe learned representations through latent-space analysis, benchmarked against exact Bayesian inference.

Machine Learning in Chaotic Systems — Apr. 2024 – Jul. 2024
Undergraduate Research

- Implemented and compared vanilla RNNs and LSTMs for multi-step forecasting of simulated chaotic triple-pendulum trajectories.
- Built training and evaluation pipelines on Imperial's HPC GPUs, investigating forecasting limitations from nonlinear dynamics and sensitivity to initial conditions.

Hobbies and Interests

- Rowed for Imperial College at **Henley Royal Regatta** in the Prince Albert Challenge Cup
- Demonstrator in technical skiing videos used for **Austrian National Skiing** instructor program
- Fluent in **English**, **German** and **Italian**